

FORM TP 9069

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE
EXAMINATION

MATHEMATICS

Paper 2 – General Proficiency

2 hours 40 minutes

1. Answer ALL questions in Section I, and any TWO in Section II.
2. Begin the answer for each question on a new page.
3. Full marks may not be awarded unless full working or explanation is shown with the answer.
4. Mathematical tables, formulae and graph paper are provided.
5. Slide rules and silent electronic calculators may be used EXCEPT in questions where their use is expressly forbidden.
6. You are advised to use the first 10 minutes of the examination time to read through this paper and to plan your answers. Writing may begin during this 10-minute period.

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SECTION I

Answer ALL the questions in this section.

1. Calculators, slide rules and mathematical tables must NOT be used to answer this question. Show ALL steps clearly.

- (a) Calculate the exact value of

$$\begin{array}{r} 4\frac{1}{3} - 1\frac{5}{6} \\ 1\frac{3}{7} \times 1\frac{2}{3} \end{array}$$

(3 marks)

- (b) Calculate the value of

$$\frac{0.023}{0.351}$$

giving your answer correct to 2 significant figures.

(3 marks)

- (c) (i) Some years ago, US\$1.00 (one United States dollar) was equivalent to J\$3.50 (three dollars and fifty cents, Jamaican currency). Calculate the amount in US currency that was equivalent to J\$8400.

- (ii) After devaluation J\$1.00 was worth 70% of its original value. Calculate the new rate of exchange for US\$1.00 and hence calculate the amount of Jamaican dollars that would be equivalent to US\$2400.

(7 marks)

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2. Consider the following three statements:

1. Some students play basketball.
2. Tall students are over 2 metres in height.
3. All basketball players are tall students.

- (a) Represent the statements in a suitable Venn diagram, showing and stating an appropriate universal set.

- (b) Show on your Venn diagram that

- (i) Nina is 1.5 m tall

- (ii) Robert, who is 2.2 m tall, does not play basketball.

(5 marks)

3. Solve the following equations for x :

- (a) $3(x + 2)^2 = 7(x + 2)$ (5 marks)

- (b) $9^{2x} = \frac{1}{27}$ (4 marks)

- (c) $3x = y$, given that $\sin y = \frac{\sqrt{3}}{2}$ where $0^\circ < y < 180^\circ$. (4 marks)

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4. (a) (i) Using a scale of 1 cm to 1 unit on each axis, draw on graph paper triangle ABC whose vertices are $A(2, 2)$, $B(5, 2)$ and $C(2, 4)$.
- (ii) $\triangle A'B'C'$ is the image of $\triangle ABC$ under an enlargement, centre $(0, 0)$ and scale factor $k = 2$. Draw $\triangle A'B'C'$, and state the coordinates of the vertices.
- (iii) $\triangle A''B''C''$ is the image of $\triangle ABC$ under a transformation T . The vertices are $A''(-2, -4)$, $B''(-5, -4)$ and $C''(-2, -2)$. Give a full geometrical description of the transformation T .
- (9 marks)

5. (a) Given that $f(x) = 2 \sin \frac{x}{3}$, copy and complete the following table:

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	π
$f(x)$		0.35	0.52	0.68		1.29		

- (b) Using a scale of 2 cm to represent $\frac{\pi}{12}$ on the x -axis and 10 cm to represent 1 unit on the $f(x)$ axis, draw the graph of the function $f(x)$ for the domain $0 \leq x \leq \pi$.
- (c) By drawing a suitable tangent, determine the gradient of the curve at $x = \frac{\pi}{2}$.
- (10 marks)

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6. (a) $A = \begin{pmatrix} 3 & 5 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 6 & 4 \\ -1 & -2 \end{pmatrix}$.

- (i) Calculate the matrix product AB .

(ii) If $C = \begin{pmatrix} x & -8 \\ 4 & y \end{pmatrix}$ and $2A + C = AB$,

calculate the values of x and y .

(8 marks)

- (b) The ratio of the prices of two different sheets of glass is 2 : 5. The total bill for 20 sheets of the cheaper glass and 10 sheets of the more expensive one is \$1080. If d dollars represent the cost of one sheet of the cheaper glass, determine

- (i) an expression in d for the cost of ONE sheet of the more expensive glass
- (ii) the value of d
- (iii) the cost of ONE sheet of the more expensive glass.

(6 marks)

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7. (a) The frequency distribution of the lengths of 100 rods measured to the nearest mm is given in the following table.

Using an assumed mean of 222 mm, calculate the mean length of all the rods.

Length (mm)	Frequency
200 – 204	3
205 – 209	7
210 – 214	11
215 – 219	14
220 – 224	25
225 – 229	23
230 – 234	11
235 – 239	6

(7 marks)

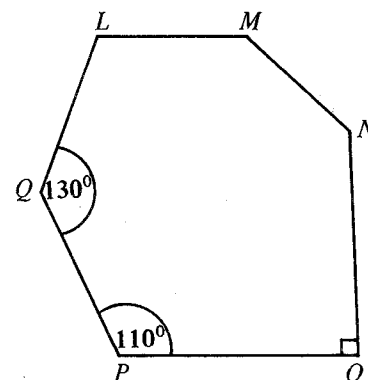
- (b) A two-digit number in base ten is formed by choosing the tens digit from the set $\{4, 5, 6, 7, 8\}$ and the units digit from the set $\{1, 2, 3, 9\}$.

- Determine the total number of different two-digit numbers that may be formed.
- Calculate the probability that the resulting number is even.
- Hence, deduce the probability that the resulting number is odd.

(5 marks)

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8.



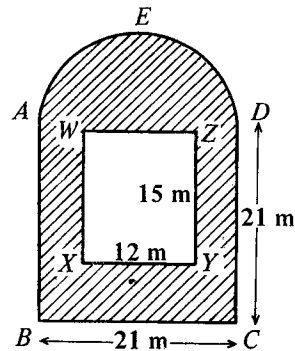
- (a) $LMNOPQ$ is a hexagon (not drawn to scale) with $\hat{Q} = 130^\circ$, $\hat{P} = 110^\circ$ and $\hat{O} = 90^\circ$. $\hat{L} = \hat{M} = \hat{N}$.

- (i) Calculate the value of $\hat{L}$.

- (ii) Given that $PO = 4$ cm and the area of $\triangle NOP$ is 12 cm^2 , calculate the length of PV in cm, giving your answer correct to one decimal place.

(7 marks)

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(b) (Take π as $\frac{22}{7}$)

The diagram above, **not drawn to scale**, represents a plot of land $ABCDE$ in the shape of a square of sides 21 m with a semicircle at one end.

- (i) Calculate in metres, the perimeter of the plot of land.
- (ii) $WXYZ$ is a rectangular flower bed of length 15 m and width 12 m. Calculate in square metres, the area of the shaded section.
- (iii) The soil in the flower bed is replaced to a depth of 30 cm. Calculate, in cubic centimetres, the volume of the soil replaced, writing your answer in standard form.

(7 marks)

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SECTION II

Answer any TWO questions in this section.

RELATIONS AND FUNCTIONS

9. (a) Copy and complete the following table for the function $f : x \rightarrow \frac{1}{x^2}$ for real values of x in the domain $0.5 \leq x \leq 3.5$.

x	0.5	1.0	1.25	1.5	2.0	2.5	3.0	3.5
$f(x)$		1.0	0.64			0.16		0.08

- (b) Using a scale of 4 cm to represent 1 unit on the x -axis and 2 cm to represent 0.1 unit on the $f(x)$ axis, draw the graph of the function $f : x \rightarrow \frac{1}{x^2}$ for real values of x in the domain $1 \leq x < 4$.
- (c)
 - (i) Using the same scale and axes, draw the graph of the function $g : x \rightarrow \frac{x}{10}$ for the same domain.
 - (ii) Use your graph to solve the equation $\frac{1}{x^2} = \frac{x}{10}$ for real values of x such that $1 \leq x < 4$.
 - (iii) Hence, deduce one solution of the equation $64x^3 = 10$ for real values of x .

(15 marks)

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10. At a factory, two types of motor cycles, M_1 and M_2 are assembled. The factory can produce a maximum of 80 motor cycles daily. For each M_2 motor cycle made, there must be at least two M_1 motor cycles produced. It takes one workman to produce each M_1 motor cycle but three workmen to produce each M_2 motor cycle. Factory policy requires that at least 90 workmen be employed.

Let x represent the number of M_1 motor cycles and
 y the number of M_2 motor cycles produced daily.

- (a) Write down **THREE** inequalities, not including $x \geq 0$, $y \geq 0$, which represent the above conditions.
- (b) Using a scale of 2 cm to 10 motor cycles on EACH axis, **shade, on graph paper**, the region which satisfies the inequalities.
- (c) Given that a profit of \$250 is made on each M_1 motor cycle and \$200 on each M_2 motor cycle, use your graph to determine the values of x and y which give the maximum profit.

Calculate the maximum profit.

(15 marks)

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TRIGONOMETRY

11. The foot, F , of a hill and the base B , of a vertical tower TB , 21 metres tall, are on the same horizontal level. From the top, T , of the tower, the angle of depression of F is 36.9° . P is a point on the hill 24.5 m away from F along the line of greatest slope. T , B , F and P all lie in the same vertical plane. The angle of depression of P from T is 19.4° .

- (a) Draw a sketch to represent the information given above.
- (b) Show that
- (i) TF is 35 m approximately
- (ii) $\sin \hat{TPF} = \frac{3}{7}$ approximately.
- (c) Calculate
- (i) the gradient of the hill
- (ii) the height, in metres, of P above F , giving your answer correct to one decimal place.

(15 marks)

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12. (a) A helicopter is set on a course of 095° . A wind is blowing from a direction 50° east of north at a speed of 40 kmh^{-1} .

Using a scale of $1 \text{ cm to } 20 \text{ kmh}^{-1}$, determine by **drawing**,

- the track and ground speed of the helicopter when it has an air speed of 100 kmh^{-1}
- the course which must be set if the helicopter is to fly on a track of 095° .

(8 marks)

- (b) In an experiment, the distance d of a moving object from its starting point is given by the equation

$$d = \sqrt{3} - 2 \cos \theta,$$

where the angle θ is measured in radians.

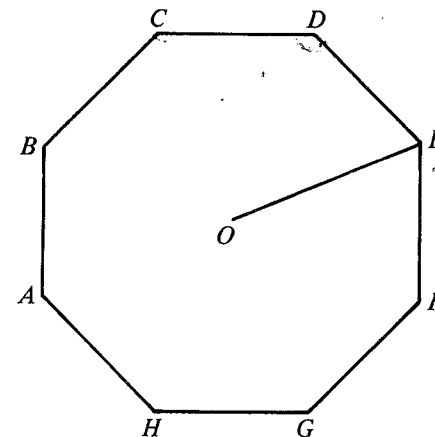
- For $-2\pi \leq \theta \leq 2\pi$, determine all the values of θ for which $d = 0$.
- State the value of d when $\theta = \frac{4\pi}{3}$, giving your answer in surd form.
- State the largest possible value of d , giving your answer in surd form.

(7 marks)

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VECTORS AND MATRICES

13.



The diagram above, **not drawn to scale**, shows a regular 8-sided polygon $ABCDEFGH$, centre O . $OE = \ell$.

- Show that $\angle COE = 90^\circ$.
- Calculate
 - the angle CDE
 - the length of CE in terms of ℓ .
- Given that P is the midpoint of CE , show that $OP = PE$.
- Given that $\underline{r}$ represents the unit vector in the direction $\vec{OD}$, and $\underline{s}$ is the unit vector in direction $\vec{OF}$, express, in terms of $\underline{r}$ and $\underline{s}$
 - $\vec{OE}$
 - $\vec{OG}$.

(15 marks)

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14. (a) Four points on a plane have the following coordinates:

$$O(0, 0); P(0, 12); Q(6, 0); R(0, -3).$$

The triangle QOR is mapped onto triangle QOP by a combination of two transformations – a reflection and a stretch.

- (i) Describe fully the transformations and determine the matrix for

(a) the reflection

(b) the stretch

(c) the combination of the two transformations.

- (ii) Determine the matrix which maps triangle QOP onto triangle QOR .
(8 marks)

- (b) The vertices of triangle ABC have coordinates $A(2, 3)$; $B(4, 6)$; $C(6, 3)$ and the triangle $A'B'C'$ has vertices $A'(3, 3)$; $B'(6.5, 6)$; $C'(7, 3)$. $A'B'C'$ is the image of ABC under a shear by a positive scale factor.

Determine, for the shear,

- (i) the equation of the invariant line

- (ii) its scale factor.

(7 marks)

END OF TEST